

HTTP Protocol
Specification

for

TMR Current Sensing Terminal

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Date	Version	Memo
June 18, 2026	V0.0.1	Initial Version
June 25, 2026	V0.0.2	Add more GET methods

Preface

This comprehensive specification is drafted to align software developers, hardware engineers and system integrators on end-to-end workflows for TMR Current Sensing terminals. All data retrieval operations are implemented via the standard HTTP GET method. Important note for the current development phase: only unsecure HTTP is supported, HTTPS is not available for use right now, and no authentication steps are needed to get valid data responses.

Currently supported GET list (Will extend if needed)

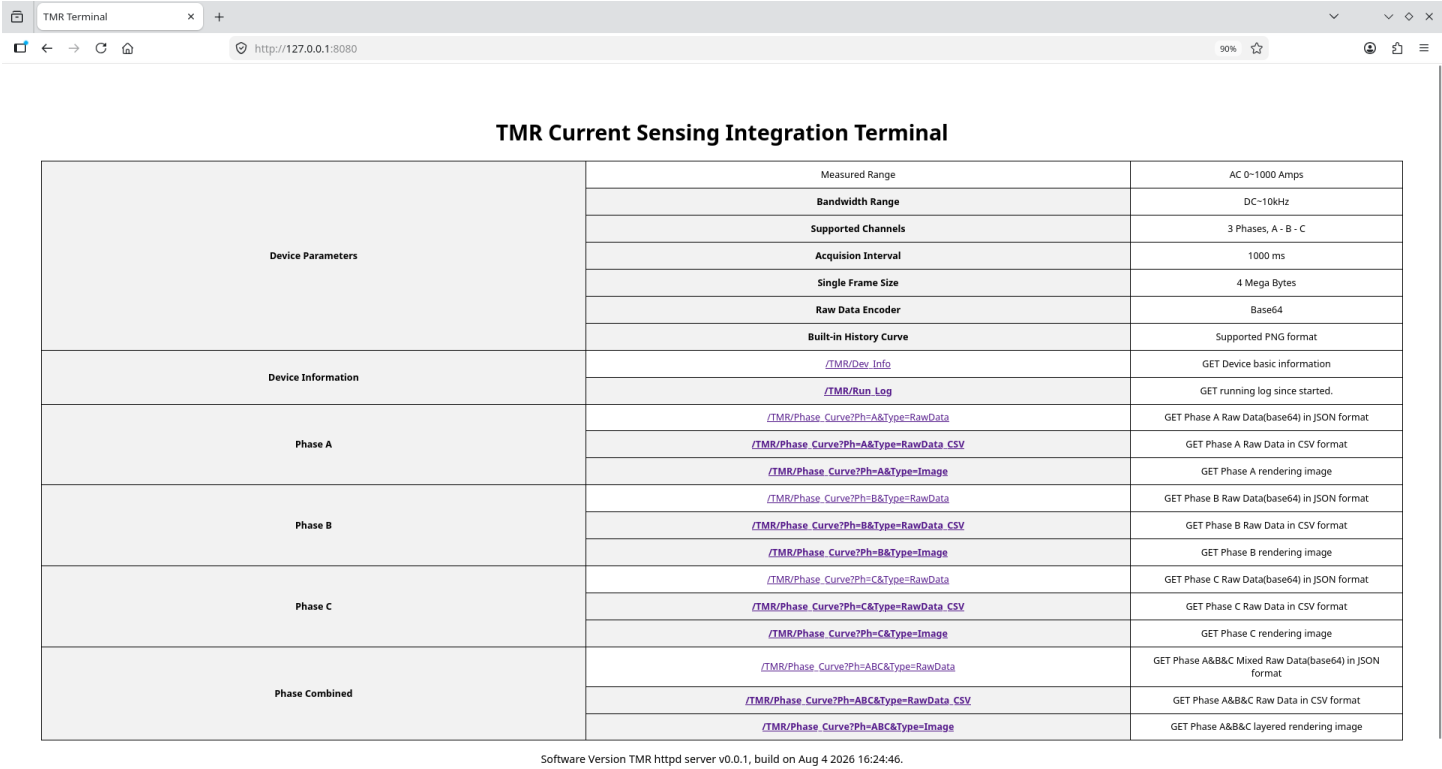
URL	Description
/index.html	The home page.
/TMR/Dev_Info	Get basic information of the terminal.
/TMR/Run_Log	Get running log since starts.
/TMR/Phase_Curve?Ph=A&Type=Image	Get Phase A rendering curve image.
/TMR/Phase_Curve?Ph=A&Type=RawData	Get Phase A raw data.
/TMR/Phase_Curve? Ph=A&Type=RawData_CSV	Get Phase A raw data in CSV format.
/TMR/Phase_Curve?Ph=B&Type=Image	Get Phase B rendering curve image.
/TMR/Phase_Curve?Ph=B&Type=RawData	Get Phase B raw data.
/TMR/Phase_Curve? Ph=B&Type=RawData_CSV	Get Phase B raw data in CSV format.
/TMR/Phase_Curve?Ph=C&Type=Image	Get Phase C rendering curve image.
/TMR/Phase_Curve?Ph=C&Type=RawData	Get Phase C raw data.
/TMR/Phase_Curve? Ph=C&Type=RawData_CSV	Get Phase C raw data in CSV format.
/TMR/Phase_Curve?Ph=ABC&Type=Image	Get Phase A+B+C layered rendering curve image.
/TMR/Phase_Curve?Ph=ABC&Type=RawData	Get Phase A+B+C mixed raw data.

/index.html

Description

Return the home page of the device. In this page, it displays the basic performance parameters of the terminal.

Screenshot

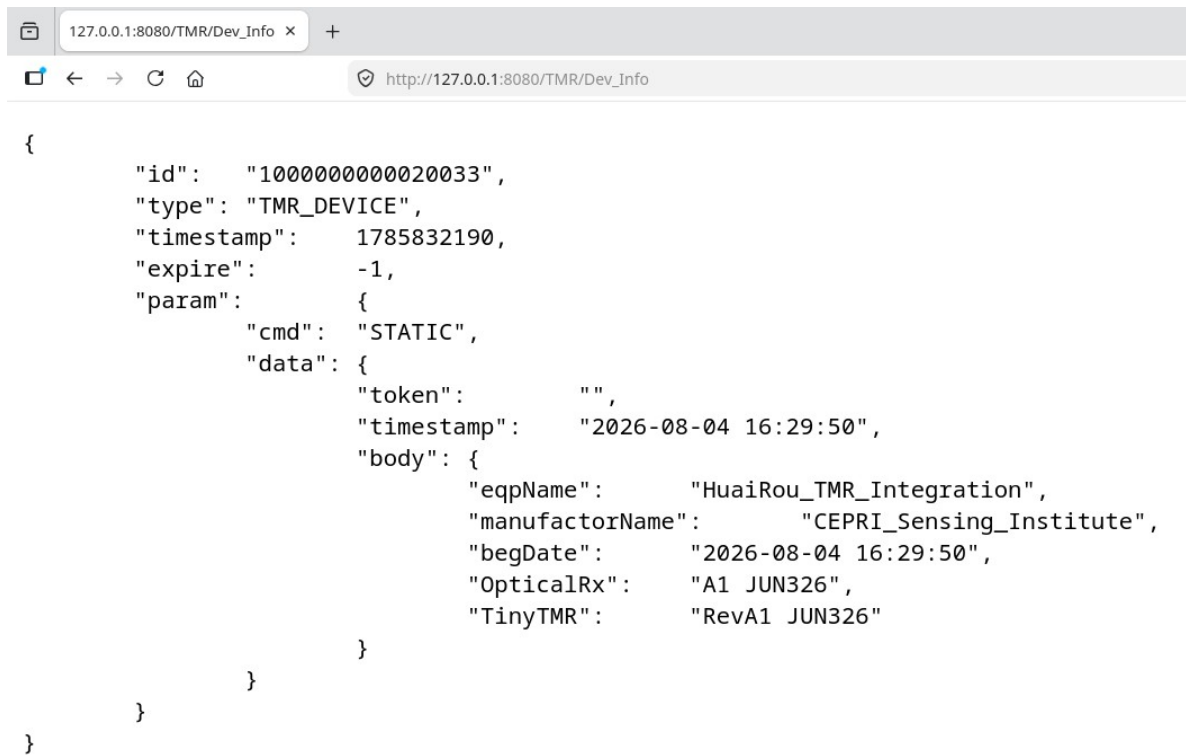


/TMR/Dev_Info

Description

To view the basic terminal details, issue a GET request to /TMR/Dev_Info, which returns a summary table of static device information.

Screenshot

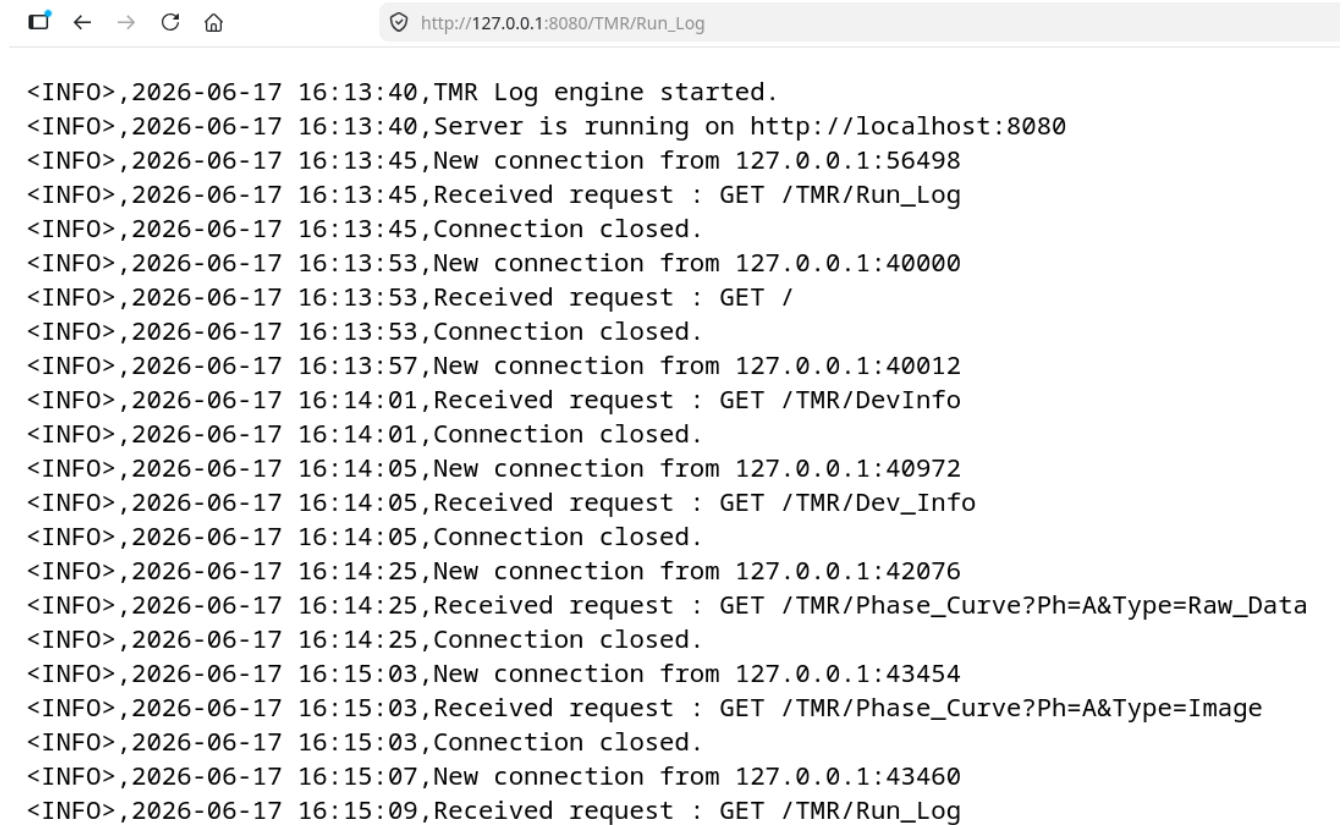


/TMR/Run_Log

Description

Send a GET request to /TMR/Run_Log to retrieve the full running log generated since device startup. The endpoint is designed to assist in rapid troubleshooting when errors occur.

Screenshot



The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:8080/TMR/Run_Log`. The main content area displays a log of server events in a monospaced font. The log entries are as follows:

```
<INFO>,2026-06-17 16:13:40,TMR Log engine started.
<INFO>,2026-06-17 16:13:40,Server is running on http://localhost:8080
<INFO>,2026-06-17 16:13:45,New connection from 127.0.0.1:56498
<INFO>,2026-06-17 16:13:45,Received request : GET /TMR/Run_Log
<INFO>,2026-06-17 16:13:45,Connection closed.
<INFO>,2026-06-17 16:13:53,New connection from 127.0.0.1:40000
<INFO>,2026-06-17 16:13:53,Received request : GET /
<INFO>,2026-06-17 16:13:53,Connection closed.
<INFO>,2026-06-17 16:13:57,New connection from 127.0.0.1:40012
<INFO>,2026-06-17 16:14:01,Received request : GET /TMR/DevInfo
<INFO>,2026-06-17 16:14:01,Connection closed.
<INFO>,2026-06-17 16:14:05,New connection from 127.0.0.1:40972
<INFO>,2026-06-17 16:14:05,Received request : GET /TMR/Dev_Info
<INFO>,2026-06-17 16:14:05,Connection closed.
<INFO>,2026-06-17 16:14:25,New connection from 127.0.0.1:42076
<INFO>,2026-06-17 16:14:25,Received request : GET /TMR/Phase_Curve?Ph=A&Type=Raw_Data
<INFO>,2026-06-17 16:14:25,Connection closed.
<INFO>,2026-06-17 16:15:03,New connection from 127.0.0.1:43454
<INFO>,2026-06-17 16:15:03,Received request : GET /TMR/Phase_Curve?Ph=A&Type=Image
<INFO>,2026-06-17 16:15:03,Connection closed.
<INFO>,2026-06-17 16:15:07,New connection from 127.0.0.1:43460
<INFO>,2026-06-17 16:15:09,Received request : GET /TMR/Run_Log
```

/TMR/Phase_Curve?Ph=A&Type=Image

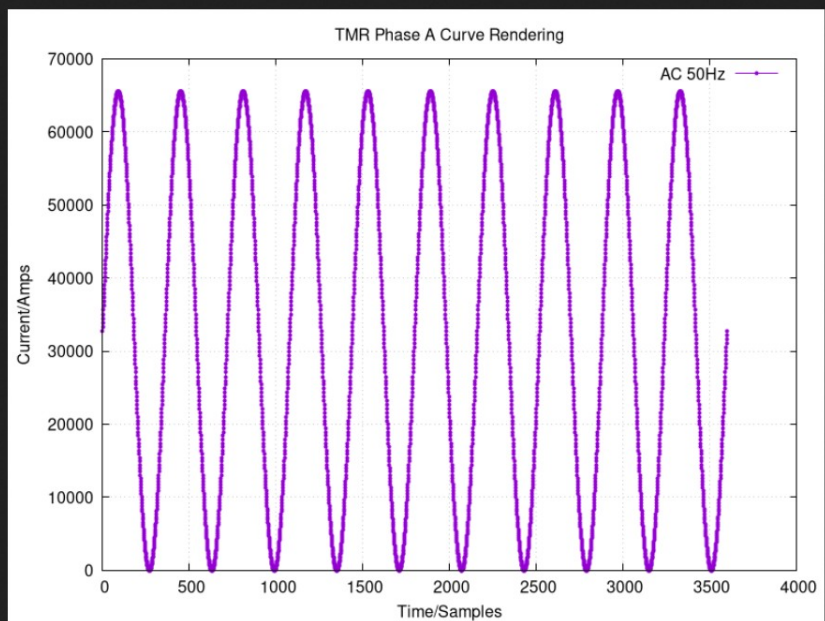
Description

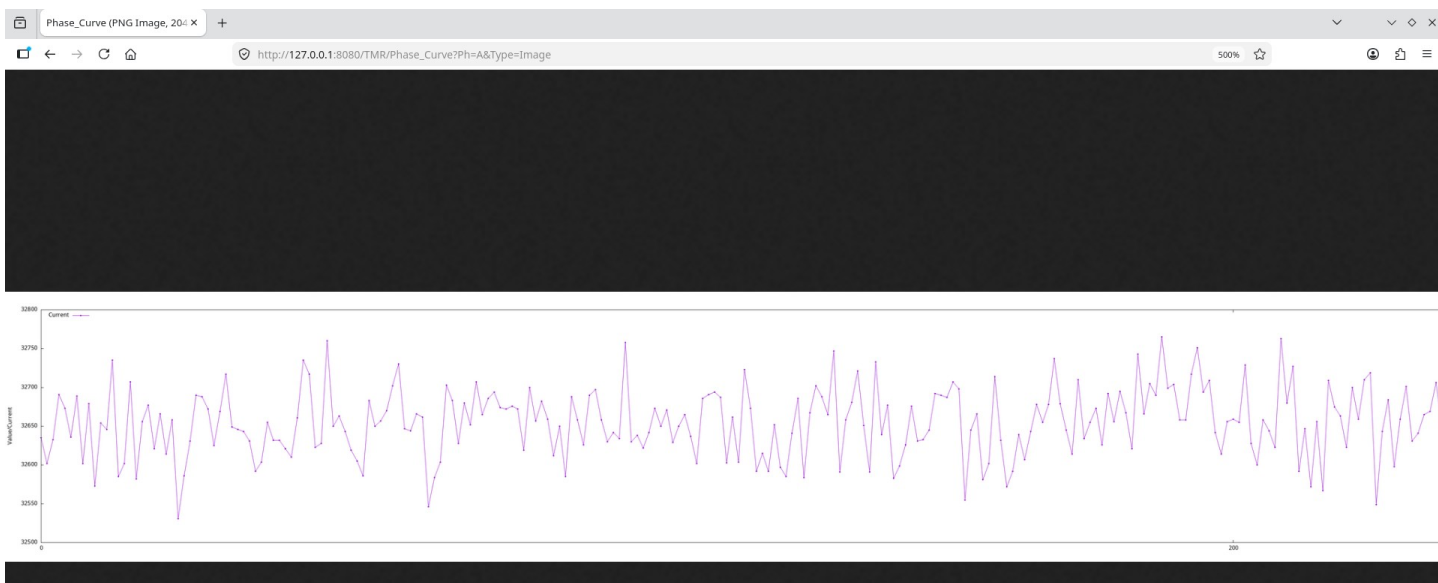
Retrieve the rendered curve image for phase A. By default, the terminal outputs the curve in PNG format, and the total number of sampling points is configurable during the deployment phase.

Screenshot



http://127.0.0.1:8080/TMR/Phase_Curve?Ph=A&Type=Image





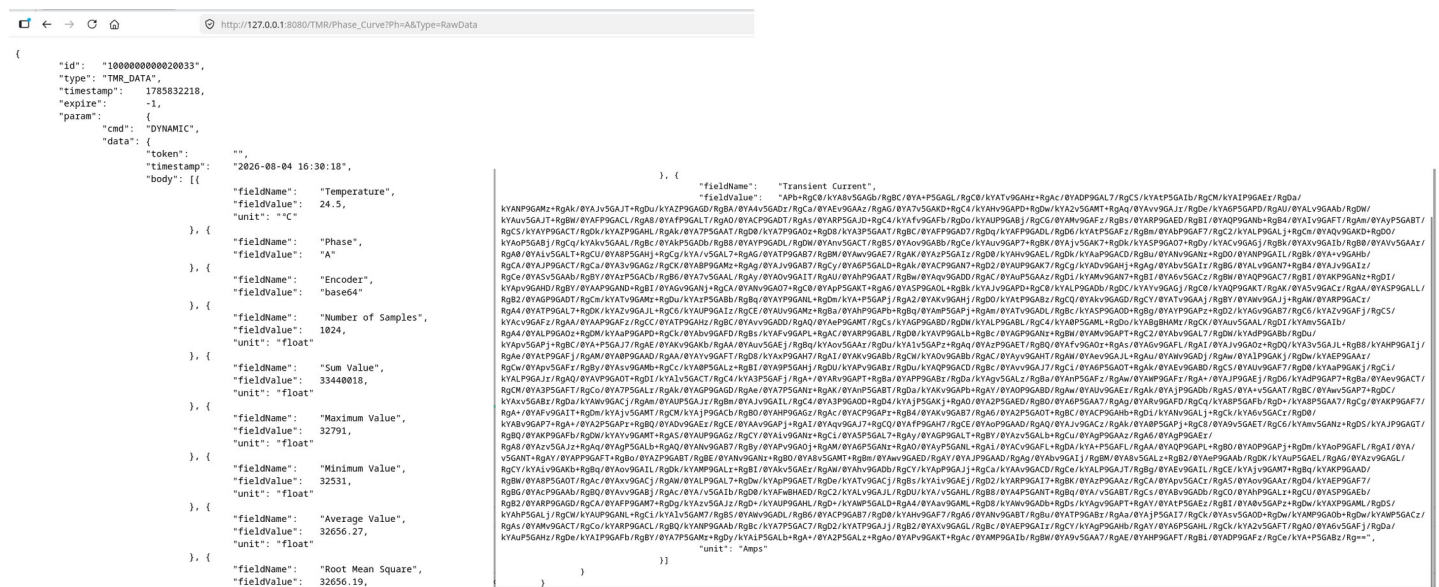
```
> file /tmp/TMR_Phase_A.dat.csv.png
/tmp/TMR_Phase_A.dat.csv.png: PNG image data, 20480 x 768, 8-bit/color RGB, non-interlaced
```

/TMR/Phase_Curve?Ph=A&Type=RawData

Description

Retrieve the raw data for phase A. The terminal acquires specified numbers of bytes of data, which is encoded via Base64 algorithm since HTML does not support binary data format. All data points are 4 bytes floating point values.

Screenshot

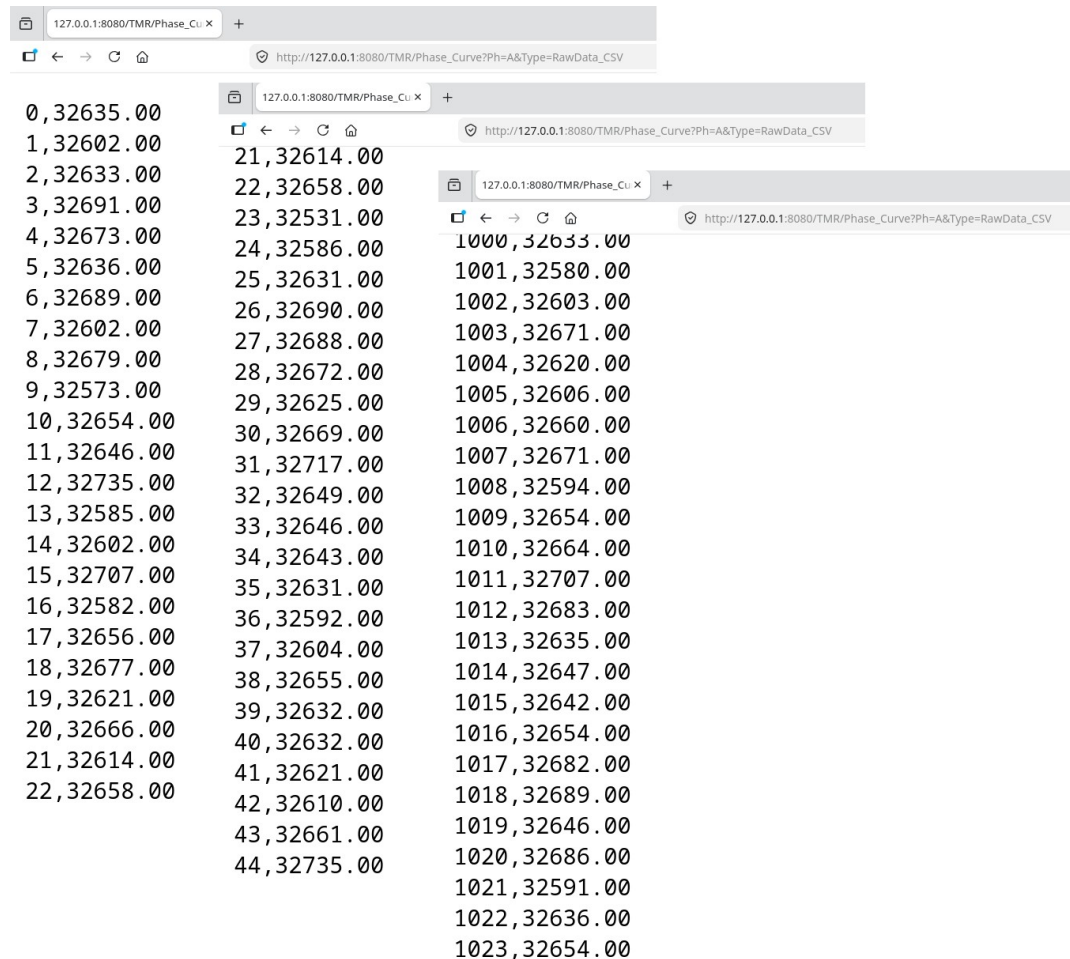


/TMR/Phase_Curve?Ph=A&Type=RawData_CSV

Description

Retrieve the CSV (Comma-Separated Values) raw data for phase A. This is used to verify the base64 decoder's functionality. These data also facilitates curve plotting in Excel, making it a valuable resource during the debugging stage.

Screenshot



0,32635.00	21,32614.00	1000,32633.00
1,32602.00	22,32658.00	1001,32580.00
2,32633.00	23,32531.00	1002,32603.00
3,32691.00	24,32586.00	1003,32671.00
4,32673.00	25,32631.00	1004,32620.00
5,32636.00	26,32690.00	1005,32606.00
6,32689.00	27,32688.00	1006,32660.00
7,32602.00	28,32672.00	1007,32671.00
8,32679.00	29,32625.00	1008,32594.00
9,32573.00	30,32669.00	1009,32654.00
10,32654.00	31,32717.00	1010,32664.00
11,32646.00	32,32649.00	1011,32707.00
12,32735.00	33,32646.00	1012,32683.00
13,32585.00	34,32643.00	1013,32635.00
14,32602.00	35,32631.00	1014,32647.00
15,32707.00	36,32592.00	1015,32642.00
16,32582.00	37,32604.00	1016,32654.00
17,32656.00	38,32655.00	1017,32682.00
18,32677.00	39,32632.00	1018,32689.00
19,32621.00	40,32632.00	1019,32646.00
20,32666.00	41,32621.00	1020,32686.00
21,32614.00	42,32610.00	1021,32591.00
22,32658.00	43,32661.00	1022,32636.00
	44,32735.00	1023,32654.00

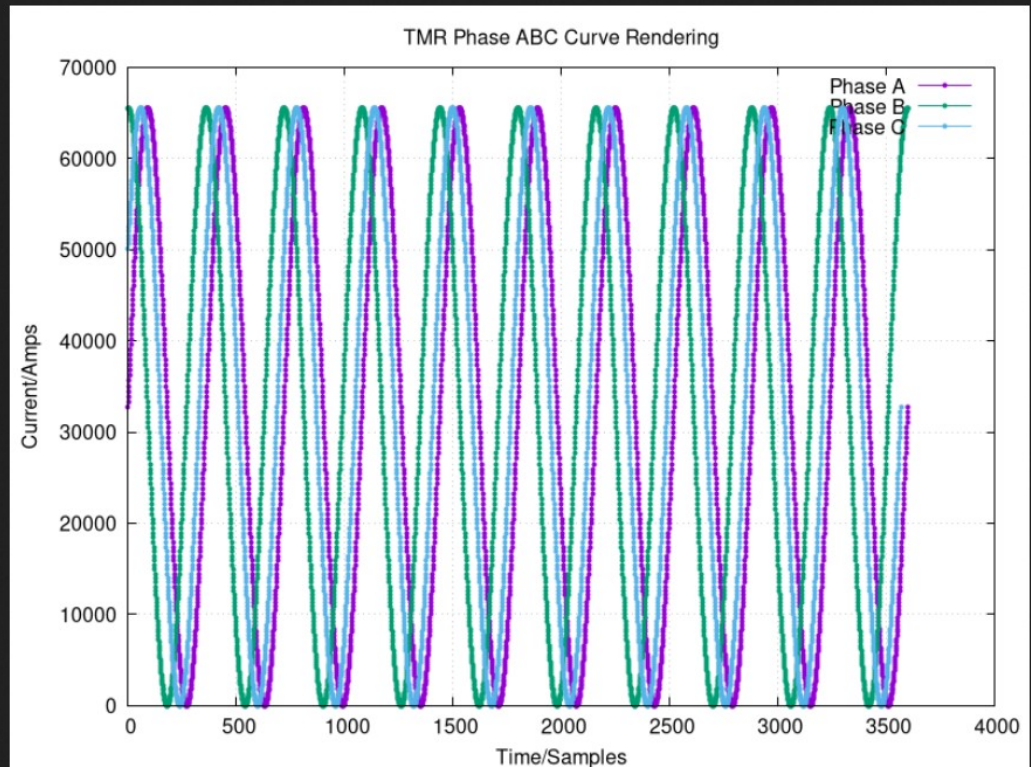
/TMR/Phase_Curve?Ph=ABC&Type=Image

Description

Retrieve the combined rendered curve image for phase A, B and C. By default, the terminal outputs the curve in PNG format, and the total number of sampling points is configurable during the deployment phase.

Screenshot

http://127.0.0.1:8080/TMR/Phase_Curve?Ph=ABC&Type=Image

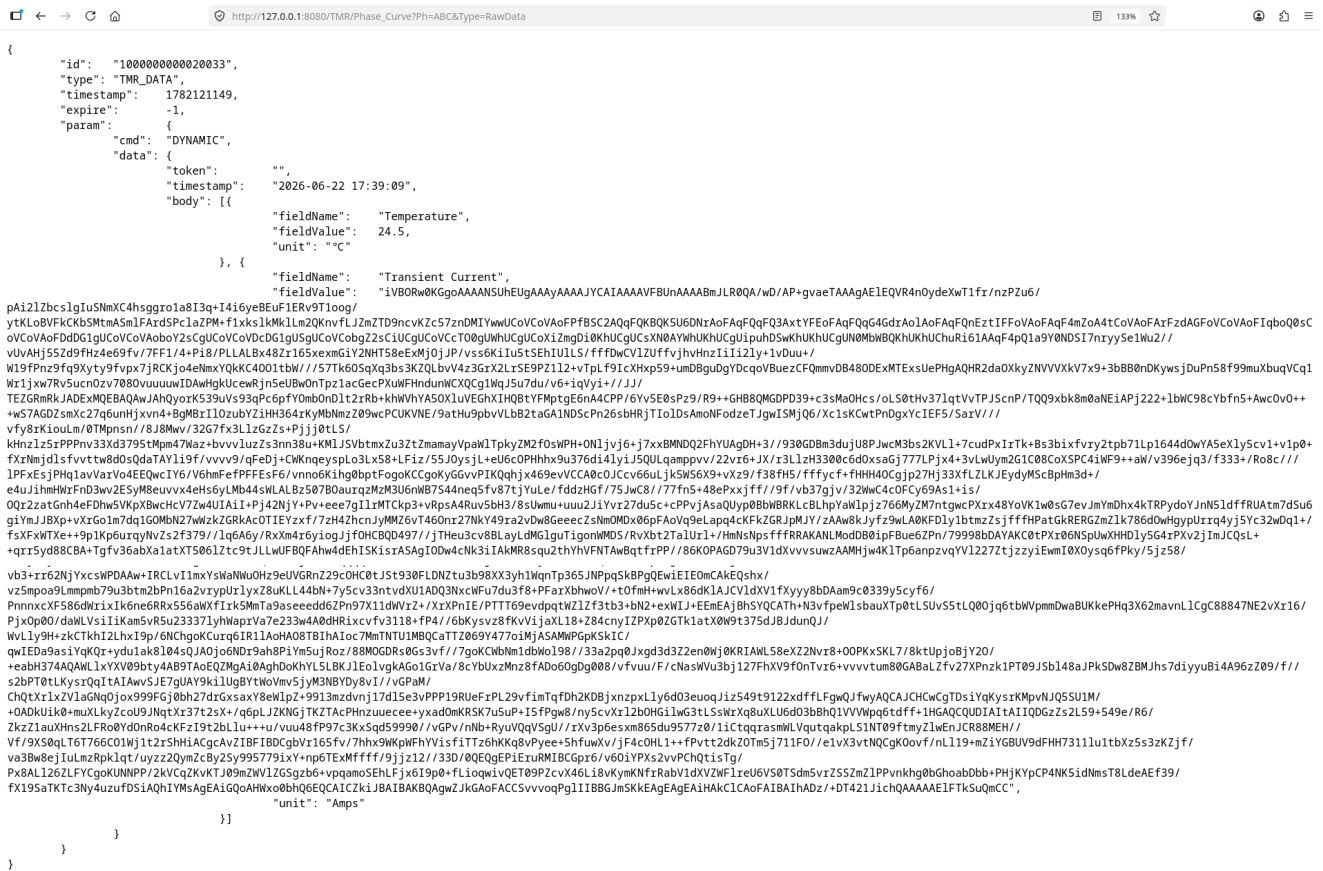


/TMR/Phase_Curve?Ph=ABC&Type=RawData

Description

Retrieve the raw data for phase A, B and C. The terminal acquires specified numberss of bytes of data, which is encoded via Base64 algorithm since HTML does not support binary data format. All data points are 4 bytes floating point values.

Screenshot



```
{
  "id": "100000000020033",
  "type": "TMR_DATA",
  "timestamp": 1782121149,
  "expire": -1,
  "param": {
    "cmd": "DYNAMIC",
    "data": {
      "token": "",
      "timestamp": "2026-06-22 17:39:09",
      "body": [
        {
          "fieldName": "Temperature",
          "fieldValue": 24.5,
          "unit": "°C"
        },
        {
          "fieldName": "Transient Current",
          "fieldValue": "iVBORw0KGgoAAAANSUHEuGAAyAAAAJCAIAAAAVFBUAAAABJLR0QA/wD/AP+gvaeTAAAgAEIEQVR4nOydeXwT1fr/nzPZu6/
          pAi2lZbcs1gIu5MmX4hsggro1a8I3q+14i6yEuf1ERv9T1oog/
          ytkLOBVfCKbSMtmASm1Fard5SPc1aZPw+f1xks1MkMlLn2QKnvFLJZmZTD9ncvKzC572ndMIYwUoCoVaOPfBSC2AQqFQBKQSU6DNrAoFAqFQqF03AxtYFEoFAqFQqG4GdZrAo1AoFAqFQnEzt1FFoVIAoFAqF4mZoA4tCoVAoFArFzdaGFoVCoVAoFIqboQ8sC
          oVCoAoFDDdG1gUoCoVaAoobyZ2sCgUoCoVCoDCG1gU5gUoCoVCoBgZ2sC1UoCgUoCoVCoT00gUwHucUoCoXiZmgDi0KhUgUoCgXN0AYwhUkHucGUpuHdSwkhUKHucGUN0MBwBQKhUKHucHuRi61AAqF4pQ1a9Y0NDS17rnyySe1Wu2//
          vUvAHj5S2d9fH24e69fV/7FF1/4+Pi8/PLLALBx48Z165xexmgGiY2NHT58eXmJ0jP/vss6KiIu5tEhIUILS/fffdwCV1ZUffvjvhVhNZ1iI2lY+1vDu+//
          W19FPnzf9qXy9tY9fvpX7JRCKj04eMmXqKkC4001tbW//577TK605qKq3bs3KZQLbv4z3GxX2LrSE9PZ112+vTlP9ICxHxp59+umDBguDgYDcqoVBuezCFQmmvDB480DEXTExsUePHgAQHR2da0XkyZNVVXkv7x9+3bBB0nDKywsjDuPn58f99muXbuqVCq1
          W1r1jxwR7v5ucnOzv7080vuuuuuIDAwHgKlucwrJn5eUBwOnTpz1acGecPXuMFHndunWCXQcG1JwQJ5u7du/v6+iqVyi+/1J1/
          TEZGRmRkJADEXMQEBAQAwJAhQyorK539uV593qC6pfY0mbOnD1t2rRb+khWVhYA50X1uVEGHXIHQ8tYFMptgE6nA4CPP/6Yv5E0sPz9/R9++GH88QMGPDP39+c3sMaOHcs/oLS0thv371qtVvTPJ5cnP/YQQ9xbk8m0aNEiAPj222+1bwMC98cYbfn5+Awc0v0++
          +w57AGDZsmXc27q6unHjxvn4+BgMBz1I0zubY2iHH364rKyMbNmzZ09wcPCUKVNE/9aThU9pbvVLb82taGA1INDScPn26sBHRTiO1DsAmonFodzeTJgwISMjQ6/Xc1sKCwtPnDgxYcIEF5/SarV//
          vfy8zKioulm/0TMpsn1/8J8Mwv/32G7fX3L1zGzZs+Pjj0tLS/
          kHnzLz5zPPPNv33Xdz3795tMpe47Waz+bvuvluzZ3nn38u+KMJJSVbtmxZu3ZtZmamayVpaW1LtpkyZM2f0sWPH+ON1jvj6+7j7xBMNDQ2FhYUagOH+3//930GDBm3dujU8PJwcM3bs2KVL1+7cudPxIrTk+B53b1xfvry2tpb71Lp1644d0wYA5eXly5cv1+v1p0+
          fXrNmjd1sFvvtw8d0sQaTAY1l19f/vvvv9/qeDj+CWknqesyLo3Lx58+Lfiz/55J0ysjL+eU6cOPHhhx9u376d14lyiJ5QULqamppvv/22vr6+JX/r3LlZH3300c6d0xsaGj777LPJx4+3vLwUym2G1C08CoXSPC4iWF9++aw/v396eq3/f333+/Ro8c///
          1PFxEsjPHq1avVarVo4EEQwcIY6/V6hmFefPFfEsF6/vnno6KiHg0bptFogokCCgokyGgvvP1Qqkhj469evVCCA0c0Jccv66uLjK5W56X9+vXz9/f38fH5/fffyf+fHHH40Cgjp27Hj33xFLZLKJeydyM5cBphm3d+/
          e4uJihmHwRfnd3wv2E5yM8euvvx4eHs6fLmb44SWLALBz507B0auzqzMz3U6nW87544neq5fV87tjYuLe/fddzHGf/75JwC8//77fn5+48ePxxjff/9f/vb37gJv/32Ww4c0FCy69As1+is/
          QQZ2zatGnh4eFDhw5VKpXbWcHcV7Zw4UIA1I+Pj42NjY+Pv+eee7g1l1mTCKp3+vRpsA4Ruv5bH3/8sUumu+uuu2J1Yr27du5c+CPVjAsaQUpy0BbWBRKLcBLhpYaW1pjz766MyZM7ntgwcPXrx48YvK1w0sG7evJmYmDhx4kTRPydoJnN51dfFruAtm7dSu6
          giYmJJBXp+vXrGo1m7d1GOMB27WnzKZGRKAc0TIEYzx/7Z4H2cnjYmMZ6vT460nz27NkY49ra2vDw8GeeecZsNmOMDx06pFAoV9eLapq4cKFZGRJpMJY/zAAw8KJyFz9wLA0KFdLy1btmZsZjffFHPatGKRERGMZ1k786d0wHgyPuriq4yJ5Yc32WdQ1+/
          fsXFwTXtX++9p1Kp6uzqYnvZs2f379//1q6A6y/RxXm4z6yiojgJfFOHCBQD497//jTheu3cv8B8yLdMGLgu1igonIMDS/RvXbt2Ta1Uz1r/HmNsNpsffffRRAKANLModDB8ipFBue6ZPn/79998bDAYAKC0tPxR06NspuXHHdLy5G4zPXvZjImJCqSL+
          +qrx5y88BCBA+Tgfv36abXa1atXtS061Ztc9tJLwUfBQFAhw4dEH5K1srASAgIDow4cNk31IAkMR8squ2thYhVfNTAwBqtFrPP//86KOPAGD79u3V1dXvvvsuwZAAmHjw4K1TP6anpzvqV1227ZtjzzyiEwm1X0ysq6fPKy/5jz58/

          vb3+rr602NjYxcsMPDAAw+TRCLV11mxYsWmNUoH9eUvGRnZ29C0H0tJst930FLDNZtu3b98XX3yh1MqnT3p65JNPpq5k8PqQEWIEIE0mCAKEQshx/
          vZ5mpoa9Lmmpb79u3btm2bPn16a2vrypU1yxZ8uK1L44Bn+7y5cv33ntvdXU1ADQ3NxcWfUdu3f8+PFaxXbhwoV/+t0fHm+wwLx86dK1AJCV1dXV1Fxyyy8bDAam9c339y5cyf6/
          PnnnxcXF586dW1r1xIk6ne6RRx556aWXfIrK5Mnta9aseeidd6Pn97X11dWVrzZ/XrPnIE/PTT769evdpqtWZ1Zf3tb3+bN2+exWIJ+EEmEajBhSYQATh+N3vfpeW1sbauXbtp0tLSUv55tQ00jq6tWVpmmDbaWUKepHq3X62mavnL1CgC8884772VxR16/
          PjX0p00/dawLVS1I1Kam5vR5u233371jYhWapzVa7e233w4A0dHRixcvf3118+FP4//6bKysvz8fKvV1jxAL18+284cnyIZXPp0ZGTk1atX0W9t375dJBdunqJ/
          WvL1Y9H+zkCTkh12LhxI9p/6NChgoKCurq6IR11AoHA08TBiHa1oc7MmTNTU1MBQCAtTZ069Y4770iMjASAMwPGpK5k1C/
          qwIEDa9asiYqKzr+ydu1ak8104sQAJ0o6Ndr9a8h8PiVmsujRoz/88MOGDRs0Gs3vf//7GoKcWbNm1dbWo198//33a2pqq0Xgjd3d322en0Wj0KRIAWL58eXZ2Nvr8+00PKxSKL7/8ktUp0bjY20/
          +eabH374QAQWL1XXYV09btY4AB9TAoEQZMgA10aghDokHYLSBKJLEoLvgaGo1GrVa/8CybUzXmz8fADo60g0g008/vfuuu/F/cnasWvu3bj127fXV9f0nTvr6+vvvvtum80GABaLZfv27XPNzk1PT09jsb148aJPKSDw8ZBMJhs7diyyuBi4A96Z09//f//
          s2BP0tLkYsZrQ1tAIaw5JE7guAY9K1lUgB7y0wVmv5jyM3NBYD8v1//vGPmP/
          Ch0tXr1xZVlaNQoQjox999FGj0bH27dzGxax8WeLp2+9913mzdvnj17d15e3vPPP19RUEfZPL29vfimTqfHDKD8jxnzpxL1y6d30euoqJiz549t9122xdfFLFgwJfwyAQCAJCHCcgTds1YqKysrKmpvNJQ5SU1M/
          +0ADkUik0+muXkZcy0u9JnqtXr37t2sX+q6pLJZKNGeTKZTAoCPhnzuecee+yxadOmKRSK7u5uP+15fPgwb/ny5cvKr12b0HG1wG3tLS5WrxQ8uXLU6d03bBHQ1VVVWp6tdff+1HGAQQUODIA1tAI1QDgzZ5L59+549e/R6/
          Zkz21auXHns2LFRoYd0nRo4cKfZ19t2bLuu++u/vuu48fP97C3KxSqd59990//vGPv/nNb+RyuUqQV5qU//rxV3p6esxm865du9577z0/11ctqqrasmLWqutqakpLS1NT09fmyZLwEnJC888MEH//
          VVf/9X5Q0L6T766C01W112r25H1ACgcAvZ1BF1BDCgBv165fv/7Hh9WkpwFHYV1sfiTTz6hKKq8vPye+Shfuwv/Jf4cOHL1+1+Pvvt2dk20Tm5J71F10//e1vX3vtNQC9K0ovf/nL119+mZ1YGBUV9DFHH7311lU1tbXz5s3ZKJf/
          va3b08ejIuLmzRpklqt/yuzz20ymZkBy25y9957791xY+np6TEKmfFFF/9jJ212//33D/0QEQgEP1EzUrMBG6pr6/v601VPXs2vvPChQt1sg/
          Px8AL126ZLYCgQKUNPNP/2KtQcZKvKTJ09mZWU1ZG5gzb6+vpqamoSEHLFJx619pb0+flQoiqvQE09PZcv4k618vKymKfRabV1dXV2WF1reU6V50TSdm5vzZ5SzmZ1PPvnhq0bghoabDbb+PHjKjCP4NK5idNms78LdeAEf39/
          FX195aTKtC3Ny4zufD5iAQh1YMsAgE1G0AMWxoBhHQEQCA1Zk1JBAIBAKBQAgwZJkGAoFAccSvvovqPg11BBGJm5KkEAgEAgEAIHAKC1CoFAIBAIAhAdz/+DT421JichQAAAAE1FTksuQmCC",
          "unit": "Amps"
        }
      ]
    }
  }
}
```